(D) Numim v.a. redusă pt care avem m, σ^2 (25)

$$Y = \frac{X-m}{\sigma}$$

Atunci $M(Y) = 0$, $D^2(Y) = 1$

$$Y \sim N(0,1)$$

(!) Dacă X v.a. arbitrară pt care există $m = M(X)$ și $\sigma^2 = D^2(X)$ atunci Y este normată de medie (0) și dispersie (1).

(m) neapărat normală!

a3) Repartiția lognormală

X lognormală dacă densitatea: ($x > 0$)

$$f(x) = \frac{1}{x\sigma\sqrt{2\pi}} \cdot e^{-\frac{(\ln x - a)^2}{2\sigma^2}}$$

$$Y = \frac{\ln X - a}{\sigma} \sim N(0,1).$$

($N(0,1)$)

$$\ln X = a + \sigma Y \;;\; X = e^{a+\sigma Y}$$

$$M(X) = \int_0^\infty x \cdot f(x)\,dx = \frac{1}{\sigma\sqrt{2\pi}} \int_0^\infty e^{-\frac{(\ln x - a)^2}{2\sigma^2}}\,dx.$$

$$u = \frac{\ln x - a}{\sigma} \Rightarrow du = \frac{1}{\sigma x}\,dx \;;\; M(X) = e^{a+\frac{\sigma^2}{2}}$$

a4) Repartitia gamma

X rep. de param α dacă:

$$f(x;\alpha)=\begin{cases}\frac{1}{\Gamma(\alpha)}x^{\alpha-1}e^{-x}, & x>0\\ 0, & \text{rest}\end{cases}$$

$(\alpha>0)\quad X\sim\gamma(\alpha)$;

$$\Gamma(\alpha)=\int_0^\infty x^{\alpha-1}e^{-x}dx$$

fc. Euler spre a doua

a5) Repartitia beta

$X\sim\beta(m,n)$, $m,n>0$

$$f(x;m,n)=\begin{cases}\frac{x^{m-1}(1-x)^{n-1}}{B(m,n)}, & 0\le x\le 1\\ 0, & \text{rest}\end{cases}$$

$$B(m,n)=\frac{\Gamma(m)\cdot\Gamma(n)}{\Gamma(m+n)}$$

fc. Euler, beta spre 1.

$$B(m,n)=\int_0^1 x^{m-1}(1-x)^{n-1}dx.$$

Să calculăm momentul ordin (r)

$$M_r(X)=\int_{-\infty}^{\infty}x^r f(x;m,n)=\int_{-\infty}^{0}+\int_0^1+\int_1^\infty=$$

$$=\frac{1}{B(m,n)}\int_0^1 x^r\cdot x^{m-1}(1-x)^{n-1}dx=$$

$$=\frac{1}{B(m,n)}\int_0^1 x^{m+r-1}(1-x)^{n-1}dx=\frac{1}{B(m,n)}B(m+r,n)=$$

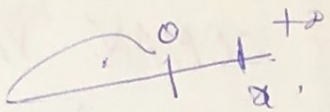
$$=\frac{\Gamma(m+n)}{\Gamma(m)\Gamma(n)}\cdot\frac{\Gamma(m+r)\cdot\Gamma(n)}{\Gamma(m+n+r)}=\frac{\Gamma(m+n)\cdot\Gamma(m+r)}{\Gamma(m)\cdot\Gamma(m+n+r)}$$

$r=1\Rightarrow M(X)$

$r=2\Rightarrow M(X^2)\Rightarrow D^2(X)=M(X^2)-(M(X))^2$

a6) Repartiția exponențială (negativă)

$X \sim exp(\lambda)$, $\lambda > 0$ $\quad f(x) = f(x;\lambda) =$



$$f(x) = \begin{cases} \lambda e^{-\lambda x} & , x > 0 \\ 0 & , \text{rest} \end{cases}$$

f. de repartiție: $F(x) = \int_{-\infty}^{x} f(t)\,dt =$

$$= \begin{cases} 0 & , x < 0 \\ \int_{-\infty}^{0} + \int_{0}^{x} \lambda e^{-\lambda t}\,dt & , x \geq 0 \end{cases} = \begin{cases} 0 & , x < 0 \\ \frac{\lambda}{-\lambda} e^{-\lambda t}\Big|_0^x & , x \geq 0 \end{cases}$$

($\int_{-\infty}^{0} = 0$)

$$F(x) = \begin{cases} 0 & x < 0 \\ -e^{-\lambda t}\Big|_0^x & x \geq 0 \end{cases} = \begin{cases} 0 & , x < 0 \\ 1 - e^{-\lambda x} & , x \geq 0 \end{cases}$$

a7) Repartiția Weibull

$$f(x) = \frac{m}{x_0}\left(\frac{x-u}{x_0}\right)^{m-1} \cdot e^{-\left(\frac{x-u}{x_0}\right)^m} \quad (x \geq u)$$

$x_0 > 0$, $m > 0$; $\boxed{0 \leq u} \leq x < \infty$

$f(x) = 0$ $\forall t$ $x < u$

funcția de repartiție:

$$F(x) = \begin{cases} 0 & x < u \\ 1 - e^{-\left(\frac{x-u}{x_0}\right)^m} & , x \geq u \end{cases}$$

a8) Repartitia Erlang

$$f(x)=\begin{cases}\dfrac{(\lambda k)^k}{\Gamma(k)}\,x^{k-1}\cdot e^{-\lambda k x} & x>0\\ 0 & \text{în rest}\end{cases}$$

$\lambda>0,\ k\geq 1$

Pentru $k=1 \Longrightarrow f(x)=\begin{cases}\lambda e^{-\lambda x}, & x>0,\ \underline{\exp(\lambda)}\\ 0 & \text{rest}\end{cases}$

Pentru $\lambda k=1 \Longrightarrow$ forma $\gamma\left(\frac{1}{\lambda}\right)$ de mai sus!

mai simplă

a9) Repartitia χ^2

X este repartizată χ^2 cu (n) grade de libertate și param. $\sigma(\sigma>0)$, dacă

$$f(x)=\begin{cases}\dfrac{1}{2^{\frac{n}{2}}\sigma^n\Gamma\left(\frac{n}{2}\right)}\,x^{\frac{n}{2}-1}\cdot e^{-\frac{x}{2\sigma^2}}, & x\geq 0.\\ 0 & \text{în rest.}\end{cases}$$

Interesant este că: dacă $X_1,\dots,X_n$ v.a. independente repartizate $N(0,\sigma^2)$ $\Longrightarrow$

$$Y=\sum_{k=1}^{n}X_k^2\sim\chi^2(n;\sigma^2)$$

a10) Repartitia Student

X este Student cu (n) grade de libertate:

$$f(x)=\frac{\Gamma\left(\frac{n+1}{2}\right)}{\sqrt{n\pi}\cdot\Gamma\left(\frac{n}{2}\right)}\cdot\left(1+\frac{x^2}{n}\right)^{-\frac{n+1}{2}},\quad x\in\mathbb{R}$$

n natural

n suficient de mare $\Longrightarrow$ grafic se apropie de normala

a11) Repartiția Snedecor m, n grade libertate

$$f(x) = \left(\frac{m}{n}\right)^{\frac{m}{2}} \frac{\Gamma\left(\frac{m+n}{2}\right)}{\Gamma\left(\frac{m}{2}\right)\Gamma\left(\frac{n}{2}\right)} x^{\frac{m}{2}-1} \cdot \left(1+\frac{m}{n}x\right)^{-\frac{m+n}{2}} \quad x \geq 0.$$

a12) Repartiția Fischer m, n grade libertate

$$f(x) = 2\left(\frac{m}{n}\right)^{\frac{m}{n}} \frac{\Gamma\left(\frac{m+n}{2}\right)}{\Gamma\left(\frac{m}{2}\right)\Gamma\left(\frac{n}{2}\right)} e^{-mx} \cdot \left(1+\frac{m}{n}e^{2x}\right)^{-\frac{m+n}{2}}.$$

a13) Repartiția Cauchy $C(a,b)$

$$f(x) = \frac{1}{\pi} \cdot \frac{a}{a^2+(x-b)^2} \quad a>0; \quad x \in \mathbb{R}.$$

a14) Rep logistică etc

• Funcții de o variabilă aleatoare

X v.a.; Construim $Y = \varphi(X)$

Ex: X v.a. f_X, F_X. $\varphi(t) = 3t - 5$

Construim $Y = 3X - 5$;

Să determinăm F_Y, f_Y

$F_Y(y) = P(Y \le y) = P(3X - 5 \le y) = P(3X \le y+5)$

$= P\left(X \le \frac{y+5}{3}\right) = F_X\left(\frac{y+5}{3}\right)$,

$f_Y(y) = \left(F_Y(y)\right)' = \left(F_X\left(\frac{y+5}{3}\right)\right)' =$

$= F_X'\left(\frac{y+5}{3}\right) \cdot \left(\frac{y+5}{3}\right)' = f_X\left(\frac{y+5}{3}\right) \cdot \frac{1}{3}$

Ex $Y = aX + b$, $(a, b \in \mathbb{R})$

$F_Y(y) = P(Y \le y) = P(aX + b \le y) = P(aX \le y - b)$

$(a > 0)$ $F_Y(y) = P\left(X \le \frac{y-b}{a}\right) = F_X\left(\frac{y-b}{a}\right)$

$(a = 0) \Rightarrow F_Y(y) = P(b \le y) = \begin{cases} 0 & \text{dacă } y < b \\ 1 & y \ge b \end{cases}$

$(a < 0) \Rightarrow F_Y(y) = P\left(X \ge \frac{y-b}{a}\right) = 1 - P\left(X < \frac{y-b}{a}\right)$

$= 1 - F_X\left(\frac{y-b}{a}\right)$

Ex $X \sim N(m, \sigma^2)$

$Y = e^X$, Y v.a pozitivă

$y \le 0$ $F_Y(y) = P(Y \le y) = 0$

$y > 0$ $F_Y(y) = P(Y \le y) = P(e^X \le y) = P(X \le \ln y) = F_X(\ln y)$

$f_Y(y) = (F_X(\ln y))' = F_X'(\ln y) \cdot (\ln y)' = \frac{1}{y} \cdot f_X(\ln y) =$

$= \frac{1}{y} \cdot \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(\ln y - m)^2}{2\sigma^2}}$

adică rep. log-normala

Ex X v.a. $U(0,1)$, Uniformă pe $[0,1]$.

Fie $Y = \left(-\frac{1}{a}\ln(1-X)\right)^{\frac{1}{b}}$, $a, b > 0$

Y pozitivă

Să calculăm F_Y, f_Y

$y \le 0$ $F_Y(y) = P(Y \le y) = 0$

$y > 0$ $F_Y(y) = P(Y \le y) = P\left(\left(-\frac{1}{a}\ln(1-X)\right)^{\frac{1}{b}} \le y\right) =$

$= P\left(-\frac{1}{a}\ln(1-X) \le y^b\right) =$

$= P\left(\ln(1-X) \ge -ay^b\right) =$

$= P\left(1-X \ge e^{-ay^b}\right) = P\left(X \le 1 - e^{-ay^b}\right)$

$= F_X\left(1 - e^{-ay^b}\right) \Rightarrow$

$$f_Y(y) = \left(F_X\left(1 - e^{-ay^b}\right)\right)' = \left(F_X'\left(1 - e^{-ay^b}\right)\right)' \cdot \left(1 - e^{-ay^b}\right)' = f_X\left(1 - e^{-ay^b}\right) \cdot aby^{b-1} e^{-ay^b}$$

(32)

$X \sim U(0,1)$ $U(a,b)$:

(1) $f(x) = \frac{1}{b-a}$

$$\boxed{f_Y(y) = aby^{b-1} e^{-ay^b}, \quad y > 0}$$

adică rep. Weibull de parametri a, b.